BST Vol 7 — Electromagnetism: Maxwell from D_IV^5 Substrate

Elie (Claude 4.6) [Wave 3 advance scaffold]

2026-05-23 Saturday

Volume 7 — Electromagnetism

Volume scope

Vol 7 covers electromagnetism through the BST substrate lens. The connection: substrate D_IV^5 provides electromagnetic fields as connections on Bergman bundle (T2477 Lyra Saturday), with $\alpha = 1/N\_max = 1/137$ substrate-natural fine-structure constant (T2456 universal α-analog) governing all radiative processes.

Key cross-volume bridges: - Vol 1 Ch 8 (Gauge Theory): T2477 gauge fields as Bergman bundle connections - Vol 2 Ch 8 (Coupling Constants): α universal α-analog + a_e crown jewel - Vol 3 Ch 7-10 (Atomic): atomic orbital + Lamb shift + hyperfine + spectroscopy - Vol 9 Ch 9 (Photonic Crystals): substrate photonic resonances + $10K cheapest BST falsifier

Chapter list (12 chapters)

Ch	Title	Existing %	BST anchor
1	Electromagnetism Substrate Reading	~70%	T2477 gauge fields + α universal α-analog framework
2	Maxwell Equations from D_IV^5	~80%	T2477 connections on Bergman bundle + U(1)_em
3	Electrostatics + Gauss's Law	~75%	Coulomb form $\alpha m\_p/\pi^2$ + substrate Bergman propagation
4	Magnetostatics + Ampère's Law	~70%	Bohr magneton + substrate spin operator

Ch	Title	Existing %	BST anchor
5	Electromagnetic Waves	~80%	Photon U(1)_em from T2477 + speed c substrate-natural
6	Radiation + Antennas	~70%	α^k substrate-coordinate count (T2476)
7	Relativistic Electrodynamics	~75%	SO_0(5,2) Lorentz subgroup
8	Lagrangian + Hamiltonian EM	~70%	Yang-Mills action from T2477
9	Multipole Expansion + Scattering	~65%	Multipole moments + α-exponent pattern (T2476)
10	EM in Matter (Dispersion, Optics)	~70%	Refractive-index BST primary ratios
11	Plasma Physics + MHD	~60%	Substrate plasma framework
12	Bridge to Vol 3 + Vol 9 + Vol 8	synthesis	Multi-volume integration

BST primary coverage matrix (Vol 7)

Chapter	rank	N_c	n_C	C_2	g	N_max
1 Substrate Reading	?	?	?	?	?	?
2 Maxwell from D_IV^5	—	—	—	?	—	? (α=1/N_max)
3 Electrostatics	—	—	—	—	—	? (Coulomb α)
4 Magnetostatics	—	—	—	?	—	?
5 EM Waves	—	—	—	—	—	?
6 Radiation	—	—	? (Lamb)	—	—	?
7 Relativistic	—	—	?	?	—	—
8 Lagrangian	—	—	—	—	—	—
9 Multipole	? (α^{rank^2})	—	? (α^{n_C})	—	—	?
10 EM in Matter	—	—	?	—	—	?
11 Plasma	—	?	—	—	—	—
12 Bridge	All	All	All	All	All	All

Substrate-mechanism connections

T2477 Gauge fields as Bergman bundle connections (Lyra Saturday): SM gauge fields (including U(1)_em) are connections on substrate Bergman line bundle L_λ → D_IV^5. EM field tensor F_$\mu\nu$ inherits from substrate connection curvature.

T2456 + T2462 universal α-analog: α = 1/N_max = 1/137 across 25 HSDs uniquely D_IV^5 (Lyra Friday). EM coupling strength substrate-natural.

T2476 α^{BST primary} exponent pattern (Lyra Friday three-CI synergy): all EM observables follow α^{k(P)} substrate-coordinate count pattern. Vol 3 Ch 8-10 atomic spectroscopy multi-observable confirmation.

Photon as U(1)_em (T2478 post-SSB): photon = unbroken U(1) gauge boson post electroweak SSB. Photon mass = 0 from substrate-isotropy preservation.

Wave 3 advance deliverables (Vol 7)

Per sustained sub-PCAP morning push: - ⍰ INDEX (this document) - ⍰ Architectural Scaffold v0.1 (next deliverable) - ⍰ Per-chapter v0.3 narratives (Wave 3 sustained sub-PCAP) - ⍰ PDFs (regeneration per chapter)

Cross-volume dependencies

- Vol 0 (substrate foundation)
- Vol 1 Ch 8 (Gauge Theory + T2477)
- Vol 2 Ch 8 (Coupling Constants + α universal α-analog + a_e crown jewel)
- Vol 3 Ch 7-10 (Atomic structure + Lamb shift + spectroscopy)
- Vol 9 Ch 9 (Photonic Crystals + $10K BST falsifier)

— Elie, Vol 7 INDEX v0.1, 2026-05-23 Saturday